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FPGA Implementation of FIR Filter Architecture using MCM Technology with Pipelining

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Abstract: **The transpose form of (FIR) is a pipeline structure that supports the MCM (Multiple Constant Multiplication) technology, but the direct form of the FIR filter shape is not compatible with the MCM technique. MCM is most effective for the transpose form when the overall operation is greater than the regular additions that reduce computation delays. In this paper, efficient Pipeline based FIR filter with MCM designed was implemented using Verilog HDL, simulated, synthesized in Xilinx ISE tool, and implemented on ARTIX-7 FPGA board. Results shows that area, delay and computational complexity reduced for MCM based FIR filter with pipeline comparative to without pipeline.**

Keywords – Pipeline; Verilog HDL; Artix7 FPGA; Multiple Constant Multiplication

I. INTRODUCTION

In signal processing, the FIR (Finite Impulse Response) filter is a stimulus response (or response for any period of time) filter with a short range, because it stops at zero for a limited time. In the FIR, the computation filter connects to various DSPs applications, for example, towards sound recovery, addressing, adjustable audio output and taking into account non-compliant references, such as program software defined radio etc. A great deal of attention was needed for a large set of FIR filters to meet the constraints and this filter should strengthen the high sampling rate of high-speed transmission. The increase and retention obligation amount required by the production filter will automatically increase along with the needs of the filter. A MANET crossing is limited to the capacity of the collector and none of these capacities can be restored under these difficult circumstances. There is no redundancy available in the FIR filter, real-time implementation of a large FIR filter asset Environmental limits in a large order. The filter coordinator is generally consistent and is known as Priority applications. This paper concentrates on the understanding ability to prospect the FIR filter and modify the structure and configuration using the MCM system approach along with pipeline for systematic reconfiguration.

II. LITERATURE SURVEY

Mohanty[2] proposed structure they were mainly concentrated on improving speed with increase in area. Chodoker[3] designed MCM based FIR filter to reduce the power with increase in delay. R.Mahesh[4] had concentrated on only area and power but not in delay. Menon,[5] deduce area but not concentrated on delay. J. Park[6] concentrated on parameter constrain using CMOS technology H. Chen [7] had concentrated only on power dissipation. Meher,[8] had concentrated on area rather than delay. Mamta[9] had concentrated on the IC technology not on the main constrains. In papers [10],[13,][15] concentrated on delay reduction for FIR filter. Chandrasekaran,[11] had taken a look on power and area reduction for optimized FIR filter. P. K. Meher [12] had concentrated on less area delay and memory by using DA algorithm. B.K.Mohanty,[13] has taken a 4 bit length and concentrated on energy per output and area delay products using length 4bit. R.Mahesh [14] had taken a look on the technology had used and concentrated on area rather than delay. Manish[16] had concentrated only on the technology he had used but not on the parameters constrains. Xin Lou[17] had mainly power and delay than area. Papers [18][20] had concentrated only on the technology he used what the problems had faced but not on the parameters. A.P.Vinodi,[19] had mainly concentrated on the area.

These things motivated me toward designing a FIR filter with considering improvement in area, delay and complexity of the design.

III. ESSENTIAL ALLEGIANCE OF THIS PAPER BY THE FOLLOWING

1. The number of computational design of Finite Impulse Response filter is derived by using the pipelining technique in order to reduce the computational.
2. A Block structure of transpose form FIR filter is used.
3. Designing of transpose form FIR filter using pipelining technique.
4. A low-complexity of design technique by using MCM and pipelining technique has been implemented to the FIR filter structure.

IV. ARITHMETIC EXAMINATION AND STATISTICAL SCHEME OF FIR FILTER

The product of a FIR filter of length

$$y(n) = \sum_{i=0}^{N-1} h(i) \cdot .x(n-i)$$

The FIR filter length N output can be operated via recurrence relation

$$Y(z) = [z^{-1}(.z^{-1}(z^{-1}h(N-1)+h(N-2))+h(N-3)...+h(1)+h(0)]X(z)_{V}$$

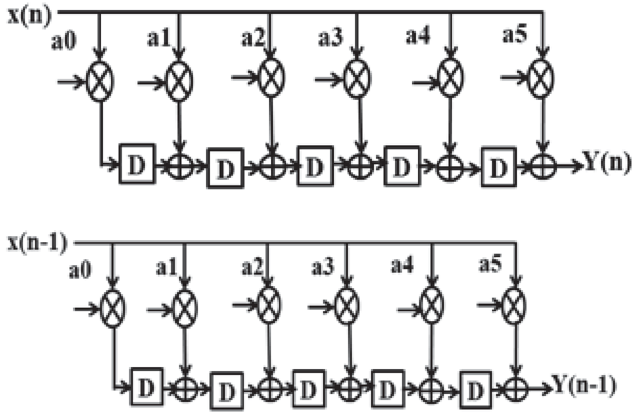

Fig. 1. FIR Filter for length N=6(a)for output y(n) (b) for output y(n-1)in this above figure multiplication, addition, and shifting operation has been done

V. EXISTING SYSTEM

The formation of FIR filter appears in the data for block size L = 4. It consists of Coefficient Selection Unit(CSU), Register Unit (RU), M number of inner product units (IPU) and one Pipeline Adder Unit (PAU). The coefficient selection unit reserves the group of the large number of channels that will be used as an implementation of the reconfigurable application. In the fixed coefficient transpose form FIR filter, the Energy per samples and Area delay products are reduced by using MCM technology. Next, we've introduced the possibility of pipeline for advance MCM structure. Pipeline is a technique for improving the overall performance of a processor.

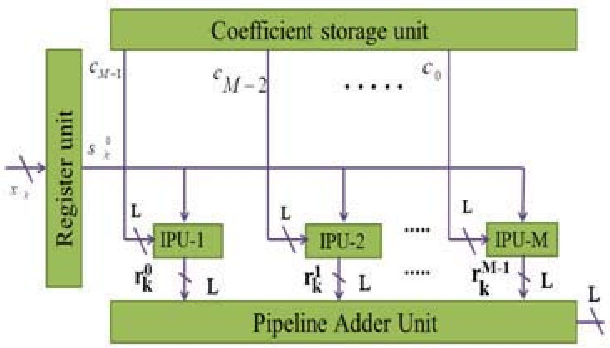

Fig. 2. Existing Structure

VI. PROPOSED FIR USING MCM TECHNOLOGY WITH PIPELINING

The basic characteristics of FIR filters are multiplication, addition and storage of symbols. Large increments should be rapid so they don't happen at all. The design is used for the multiplication combination and the delay is used to store the sample value in memory for 1 cycle per sample hour. The blocks together should have to store in a memory filter. The most commonly used filter are of two types, I.e. direct form and transpose form. Several signals come persistently and subsequently.

1. To achieve high productivity and reduction

2. Regular delay required additional pipeline registration between adders.

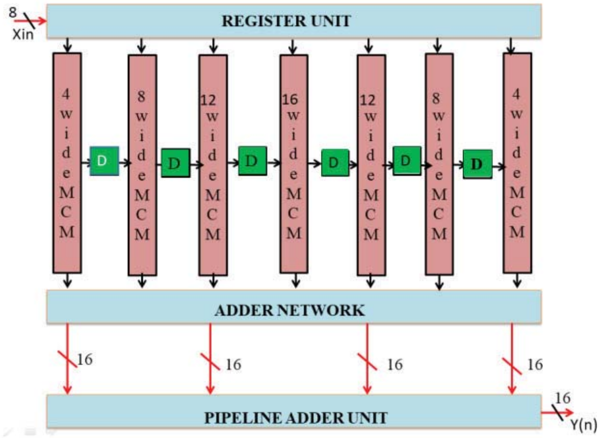

Fig. 3. Advance Structure of FIR Filter using MCM Technology with Pipeline Adder Unit

Already there is register between the adders so there is no need getting additional pipeline register. Here by using appropriate MCM operations as it is an input is multiplied by the constant number. It Provides an operating frequency to support high sampling rate .Where x (n) is the input sample, Y (n) is the output sample, h (k) is the filter coefficient, N is the filter command. Equation 1 represents the FIR N filter command as each output value is the sum of the most recent input values.

A. Register Unit

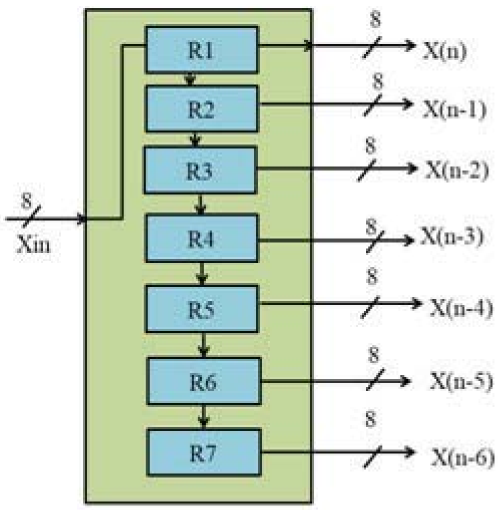

Fig. 4. FIR Filter using register unit

Here in this the register unit, where xin is an 8 bit input registers. The input will send the values parallel in parallel out from each register. So that the register output will appear as serial in serial out .where x(n) is an input it generate the output with an unit delay of time.x(n) is an input value the input value of x(n)value finally stores in register value x(n-6) by shifting 8bit the value will update to each register value I.e. from x(n-6) to x(n) the new values will be update from x(n) to x(n-6) with unit delay of time. Finally the x(n) value will be the new register value where x(n-6)will store the value what is given in x(n)

B. MCM Technology:

In any FIR filter, the multiplier is the primary constraint that defines the required filter performance. Therefore, compared to the last three decades, the inexpensive hardware design style associated with the fixed point FIR filter has been consider as the main analytical approach, as reported in the published literature. Get optimized solutions; the first coefficients are multiplied by a constant value. In the FIR filter, the multiplication operation is performed between an explicit variable (the input) and a group of constants (the coefficient). This operation can be performed using the multiple constant multiplication (MCM) followed by the aggregation of all the products.

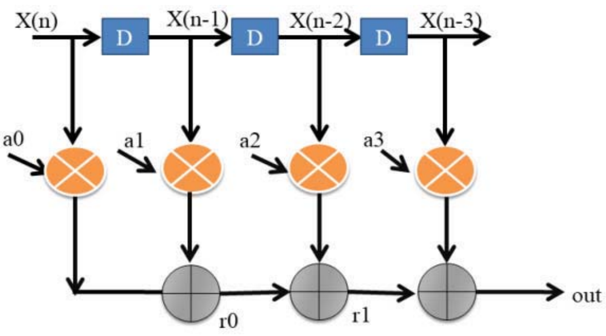

Fig. 5. Fig5: Internal Block Diagram of 4-Wide MCM

In this 4wide MCM the output of the register value are the inputs to each MCM's.In the fig5 represent the 4wide MCM. Whereas the x(n) value is the input the input value is multiple and the value is stored with an unit delay of time. likewise the new input value is generated the new value is shifted to the next register value, so that the new register value will store in the present value and the past value will be shifted to next value. Here by using MCM technology we can reduce the complexity the output value will added to the adder network from different values and then by using pipeline technique we can reduce the adder block in order to reduce the complexity. pipeline as came into the picture the output of MCM value are added to generate the output likewise each wide MCM operation is same but here we have taken only 4 constant value remaining 8,12,16,12,8,16 wide there the constant value will increase but the operation is same to all the MCM's the outputs from the register will be the input to all the MCM's after performing the operation of all MCM's each MCM will generate the different outputs from MCM's the MCM's values will be added to the adder network.

C. Adder Network:

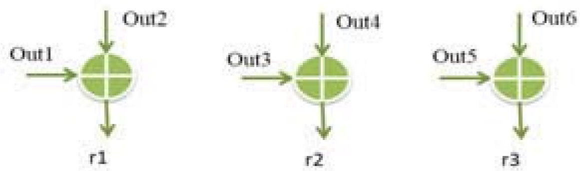

Fig. 6. Adder Network Internal Block Diagram

In this adder network the outputs of all the MCM's will be added in order to reduce the adder block that mean to reduce the complexity of design the outputs of all the MCM's will be the input of adder network .in first adder block i,e out1 is the output of 4wide MCM and out2 is an output of 8 wide MCM by adding both out1 and out2 adder has generated value i,e ―r1‖ in second block out3 is the output of 12 wide MCM out 4 is an output of 16 wide MCM by adding that both value out3 and out4 adder has generated value i,e ―r2‖ value in third block out5 is the output of 12 wide MCM out 6 is 8 wide MCM by adding those both value out5 &out6 adder has generated value i,e ―r3‖ value the out 7 value remains in adder block in order to add in pipeline adder unit to generate y(n) value.

D. Pipeline Adder Unit:

Pipeline is a significant method utilized in different applications, for example, digital signal processing systems (DSP), microchip, and so forth .It was developed from the possibility of a water pipe with nonstop water inlet without waiting that the water will emerge from the line. Therefore, it improves the speed of the basic mode in most DSP filter orders. For instance, you can accelerate or increase the use of clock strength to a similar speed in DSP systems.

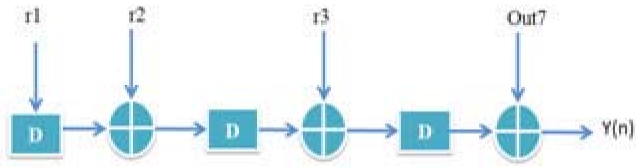

Fig. 7. PipelineAdderUnit Block Diagram

In this pipeline Adder unit the output of an adder units are the inputs of the pipeline adder networks in this pipeline adder unit the input is connected to memory the memory values are stored with an delay the input r1 value is stored in memory as q_out1. The new value q_out1 which is stored in the memory, the new value r2 is added to the q_out value which is stored in the memory unit by adding those two values the new output is generate .when new input value r3 and past output q_out values are taken from memory unit and added to adder netword and generate the output value of r3 and out7 are added and generate output y(n).

E. Theoretical Calculations of FIR Filter

TABLE I. FIR FILTER CALCULATION AT TIME T=7

Register unit	MCM outputs	Adder Network	Pipeline Adder unit
X(n) 00010010	-	-	-
X(n-1)011000101	4wide Out1 0000011001101100	Out1+out2=r1 0000011011101010	Out1+r2=q _out1 0000101100101011
X(n-2)10001101	8wide Out2 0000000111111110	Out3+out4=r2 0000010010111111	Out2+r3=q _out2 0000011011101110
X(n-3) 10001101	12wide Out3 0000001010011100	Out5+out6=r3 0000010011110000	Out3+r4=y(n) 0000101111010010
X(n-4) 01100011	16wide Out4 0000001000100011	Out7+r3=r4 0000100100110110	
X(n-5) 00001001	12wide out5 0000001010100010		
X(n-6)10000001	8wide out6 0000001001001110		
X(n-7) 00100100	4wide out7 0000010001000110		

At 7th clock pulse of time the outputs has been appeared by performing a shifting, multiplication, addition operations has been performed. Where the register values are the input to the each MCM's .The output of each MCM's the MCM's outputs has been stored by using shifting operation and then the outputs of these MCM's are added to adder network. In order to reduce the low complexity the outputs of adder units are added by combing both the MCM's output values i.e. 4wide+8wide resultsout1&out2, 12wide+16wideresultsout3 &out4, 12wide+8 wide results out5&out6 8wide+4wide results out7&r3.pipeline adder network has stored in the memory with unit with delay of time. By using pipelining technique we can reduce the area and delay units. To implement the FIR filter the coefficients of transpose form, FIR filter using pipelining reduces the over all delay area and complexity.

VII. RESULTS

In this FIR filter design is implemented on FPGA. Area and delay were analyzed by using MCM and pipelining technique using transpose form FIR filter From the analysis of existing paper it seen that the area and delay were reduced compared to direct FIR filter without using the pipeline technique.

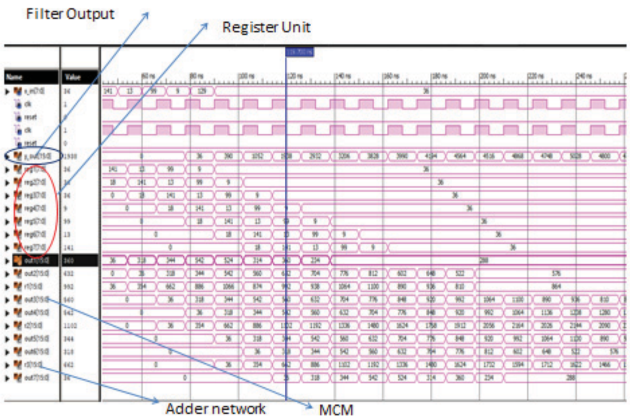

Fig. 8. register unit result

In the fig8 represent the register unit where the register unit input value are parell in parell out with an 8bit shiftregist with unit delay of time and the output values will generate the serial-in serial-out Register.

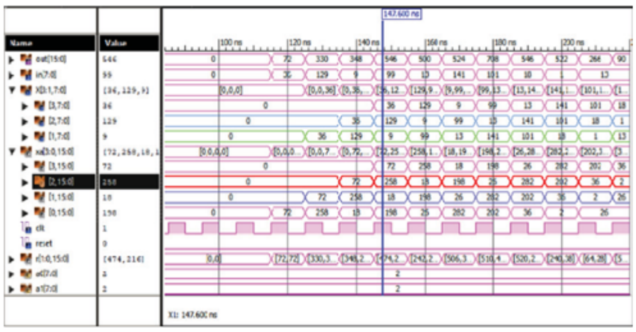

Fig. 9. 4_wide MCM technology result

In the fig9 represent 4_wide MCM technology in this technology the inputs are taken from the register unti of 4 values beacause here we have used 4-wide MCM so 4 constant values will generetate in this shifting operation multiplication operation and addition has been done to get a 4_wide output.

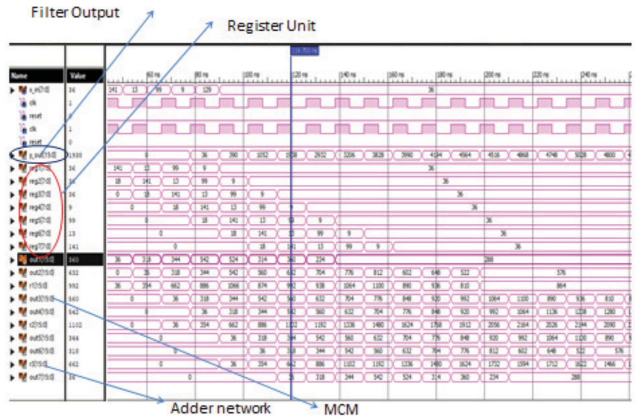

Fig. 10. FIR Filter based on MCMtechnology and Pipeline

In the above fig10 represent FIR filter based MCM technology and pipeline operation has been performed here the register unit outputs,MCM outputs,adder network output and finally pipeline has been applied to generate a actualised output here by using pipelining technique in order to reduce area and delay.

TABLE II. COMPARISON OF FIR FILTER WITH PIPELINING AND WITHOUT PIPELINING

S no	FIR Filter without using Pipeline Technology	FIR Filter using Pipeline Technology
Number of Slices	377 out of 8672	149 out of 8672
Number of Slices Flip Flops	196 out of 17344	179 out of 17344
Number of 4inputs LUTs	670 out of 17344	271 out of 17344
Number of IOs	26	98
Number of Bond IOBs	26 out of 250	98 out of 250

TABLE III. PERFORMANCE ANALYSIS OF FIR FILTER WITH PIPELINE AND WITHOUT PIPELINE

parameters	Fir filter without using pipelining	Fir filter using pipelining
Area	377 out of 8672	149 out of 8672
Delay	26.712 ns	7.902 ns

Tab1e 2 and 3 showcases the performance analysis of the FIR filter withPipeline and without pipelining. FPGA implementation of the proposed design is showned in the figure 11.

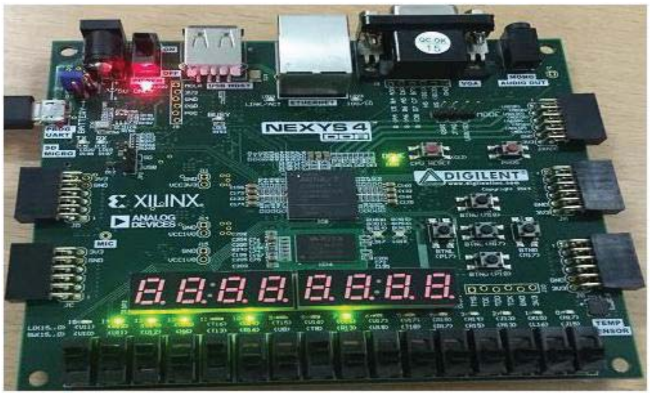

Fig. 11. FPGA Implementation of FIR Filter Architecture Using MCM Technology with Pipelining

VIII. CONCLUSION

In this paper, FIR filter design using MCM technology and pipelining technique was implemented on FPGA. The transposed FIR filter is a pipelining structure that supports the multiple constant multiplication. The implementation of the MCM technique is simpler to the transposed FIR filter, but difficult in the reconfigurable coefficients. In the fixed coefficient FIR filter, the area and delay are reduced using the MCM technique. The low complexity design using the MCM and pipelining technique was is implemented by using transposed form FIR. The FIR filter using MCM and pipelining technique structure was implemented on FPGA in order to achieve less area and delay.

REFERENCES

[1] J.G. Proakis and D. G.Manolakis, Digital Signal Processing: Principles, Algorithms and Applications. Upper Saddle River, NJ, USA: Prentice-Hall, 1996.

[2] Mohanty, Basant Kumar, and Pramod Kumar Meher. "A high-performance FIR filter architecture for fixed and reconfigurable applications." IEEE transactions on very large scale integration (VLSI) systems 24.2 (2015): 444-452.

[3] Chodoker, Trapti, Rajeev Thakur, and Puran Gour. "Multiple Constant Multiplication Technique for Configurable Finite Impulse Response Filter Design." International Journal of Computer Applications 975 (2018): 8887.

[4] R. Mahesh and A. P. Vinod, ―New Reconfigurable Architecturesfor Implementing Fir Filters With Low Complexity,‖ vol. 29, no. 2, pp. 275–288, Feb. 2010

[5] Menon, Navya V., and Sonali Agrawal. "A Speed Efficient FIR Filter forReconfigurable Applications." 2017 IEEE International Conference on Computational Intelligence and Computing Research (ICCIC). IEEE, 2017.

[6] J. Park, W. Jeong, H. Mahmoodi-Meimand, Y. Wang, H. Choo, and K. Roy, ―Computation Sharing Programmable Fir Filter For Low-Power And High-Performance APPLICATIONS,‖ *IEEE J.*Solid State Circuits, Feb. 2004. p. 1–6. K.-

[7] H. Chen and T.-D.Chiueh, ―A Low-Power Digit-Based Reconfigurable Fir Filter,‖ *IEEE* Trans. Circuits Syst. II, , vol. 53, no. 8, pp. 617–621, Aug. 2006.

[8] Meher, Pramod Kumar. "Hardware-efficient systolization of DA-based calculation of finite digital convolution." IEEE Transactions on Circuits and Systems II: Express Briefs 53.8 (2006): 707-7

[9] Mamta N. Ahuja Prachi Palsodkar "Design Of Efficient Add/Shift Algorithm For Multiple Constant Multiplication" 2015 Fifth International Conferenceon Communication Systems and Network Technologies

[10] White, Stanley A. "Applications of distributed arithmetic to digital signal processing: A tutorial review." IEEE Assp Magazine 6.3 (1989): 4-19. P. K. Meher, S.

[11] Chandrasekaran, and A. Amira, ―FPGA realizationof FIR filters by efficient and flexible systolization using distributed arithmetic,‖ *IEEE* Trans. Signal Process., vol. 56, no. 7, pp. 3009–3017,Jul. 2008.

[12] P. K. Meher, ―Hardware-efficient systolization of DA-based calculation of finite digital convolution,‖ IEEE Trans. Circuits Syst. II, Exp. Briefs, vol. 53, no. 8, pp. 707–711, Aug. 2006.

[13] B. K. Mohanty, P. K. Meher, S. Al-Maadeed, and A. Amira, ―Memory footprint reduction for power-efficient realization of 2-D finite impulse response filters,‖ IEEE Trans. Circuits Syst. I, Reg. Papers, vol. 61, no. 1, pp. 120–133, Jan. 2014.

[14] R.Mahesh and A. P. Vinod, ―A new common sub expression elimination algorithm for realizing low-complexity higher order digital filters,‖ IEEE Trans. Comput.-Aided Design Integr. Circuits Syst., vol. 27, no. 2, pp. 217–219, Feb. 2008.

[15] S. A. White, ―Applications of distributed arithmetic to digital signal processing: A tutorial review,‖ IEEE ASSP Mag., vol. 6, no. 3, pp. 4–19, Jul

[16] Manish B. Trimale Chilveri "A Review: FIR Filter Implementation" 2017 IEEE International Conference On Recent Trends InElectronics Information& Communication Technology, May 19-20, 2017, India

[17] Xin Lou, Student Member, IEEE,YaJunYu, Senior Member, IEEE,and Pramod Kumar Meher, Senior Member, IEEE "Analysis and OptimizationofProduct-Accumulation Section for Efficient Implementation of FIR Filters" IEEE Transactions On Circuits And Systems—I: Regular Papers 2016 IEEE

[18] Levent Aksoy, Student Member, IEEE, Eduardo da Costa, Paulo Flores, Member, IEEE, and José Monteiro, Member, IEEE "Exact and ApproximateAlgorithms for the Optimization of Area and Delay in Multiple Constant Multiplications" IEEE Transactions On Computer-Aided Design Of IntegratedCircuits And Systems, Vol. 27, No. 6, June 2008

[19] A. P. Vinod and E. M. Lai, ―Low power and high-speed implementation of FIR filters for software defined radio receivers,‖ IEEE Trans. Wireless Commun., vol. 7, no. 5, pp. 1669–1675, Jul. 2006.

[20] L. Aksoy, E. Costa, P. Flores, and J. Monteria.Exactand Approximate Algorithms for the Optimization of Area and Delay in Multiple Constant Multiplications. IEEE TCAD, 27(6):1013-1026, 2008. Algorithms for the Optimization of Area and Delay in Multiple Constant Multiplications. IEEE TCAD, 27(6):1013-1026, 2008.

[21] P. Cappello and K. Stieglitz. Some Complexity Issues in Digital Signal Processing. IEEE Tran. on Acoustics, Speech, and Signal Processing, 32(5):1037-1041, 1984.

[22] Georgina Binoy Joseph, R. Devanathan, "A Mathematical Framework for Online Constant Coefficient Multiplication" International Journal of Engineering and Technology Innovation, vol. 7, no. 3, 2017, pp. 217 -228

[23] Tero Rissa, Riku Uusikartano, and Jarkko Niittylahti "Adaptive FIR Filter Architectures for Run-Time Reconfigurable FPGAs" 2002 IEEE

[24] P. K. Meher, ―New approach to look-up-table design and memorybased realization of FIR digital filter,‖ IEEE Trans. CircuitsSyst. I, Reg. Papers, vol. 57, no. 3, pp. 592–603, Mar. 2010.

[25] L. Aksoy, C. Lazzari, E. Costa, P. Flores, and J. Monteria. Optimization of Area in Digit-Serial Multiple Constant Multiplications at Gate-Level. In ISCAS, to appear, 2011.